Introduction to Limits

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating limits**

Evaluate $\lim_{x\to 2} \frac{x^2-4}{x-2}$ by first simplifying the expression.

$\frac{(x-2)(x+2)}{x-2} = x+2$ for $x \neq 2$, so the limit is 4.

2 **[Foundational]** • **Target: Evaluating limits**

Evaluate by direct substitution where possible:

a $\lim_{x\to 3} (x^2+2x-1) = 14$

b $\lim_{x\to 1} \frac{x+5}{x+1} = 3$

(a) $= 14$; (b) $= 3$

3 **[Foundational]** • **Target: Evaluating limits**

Using a table of values near $x = 0$ (in radians), estimate $\lim_{x\to 0} \frac{\sin x}{x}$.

Values approach 1 from both sides; the limit is 1.

4 **[Foundational]** • **Target: One-sided limits and limits at infinity**

Evaluate each limit at infinity:

a $\lim_{x\to\infty} \frac{3x^2+5}{x^2-2} = 3$

b $\lim_{x\to\infty} \frac{2x+1}{x^2+4} = 0$

(a) $= 3$; (b) $= 0$

5 **[Foundational]** • **Target: One-sided limits and limits at infinity**

The graph of $f(x) = \frac{1}{x-2} + 3$ has asymptotes $x = 2$ and $y = 3$. Express each asymptote as a limit statement.

$\lim_{x\to 2^+} f(x) = +\infty$ (and $\lim_{x\to 2^-} f(x) = -\infty$); $\lim_{x\to\pm\infty} f(x) = 3$.

6 **[Foundational]** • **Target: More limit computations**

Evaluate $\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}$ by rationalizing the numerator.

Multiply by $\frac{\sqrt{x}+2}{\sqrt{x}+2}$: $\frac{x-4}{(x-4)(\sqrt{x}+2)} = \frac{1}{\sqrt{x}+2} \to \frac{1}{4}$.

7 **[Foundational]** • **Target: More limit computations**

Evaluate $\lim_{x\to -1} \frac{x^3+1}{x+1}$ by factoring (using the sum-of-cubes formula).

$x^3+1 = (x+1)(x^2-x+1)$, so the limit equals $\lim_{x\to -1} (x^2-x+1) = 1+1+1 = 3$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: More limit computations**

Evaluate $\lim_{x\to 0}\frac{\tan x}{x}$, using $\lim_{x\to 0}\frac{\sin x}{x}=1$ and $\tan x=\frac{\sin x}{\cos x}$.

$\frac{\tan x}{x}=\frac{\sin x}{x}\cdot\frac{1}{\cos x}\to 1\cdot\frac{1}{1}=1$.

9 **[Intermediate]** • **Target: Continuity and limits together**

A function f is defined by $f(x)=x+3$ for $x\neq 2$, and $f(2)=10$. Find $\lim_{x\to 2} f(x)$ and state whether f is continuous at $x=2$.

$\lim_{x\to 2} f(x)=2+3=5$. Since $f(2)=10\neq 5$, f is not continuous at $x=2$ — it has a removable discontinuity.

10 **[Intermediate]** • **Target: Continuity and limits together**

Estimate $\lim_{x\to\infty}\left(\sqrt{x^2+1}-x\right)$ by rationalizing.

Multiply by $\frac{\sqrt{x^2+1}+x}{\sqrt{x^2+1}+x}$: $\frac{1}{\sqrt{x^2+1}+x}\to 0$ as $x\to\infty$.

11 **[Intermediate]** • **Target: Limits from graphs and tables**

A function g satisfies $g(1.9)=5.8$, $g(1.99)=5.98$, $g(1.999)=5.998$, $g(2.1)=6.2$, $g(2.01)=6.02$, $g(2.001)=6.002$. Based on this table, estimate $\lim_{x\to 2} g(x)$.

Both sides approach 6: $\lim_{x\to 2} g(x)=6$.

12 **[Intermediate]** • **Target: Limits from graphs and tables**

Explain, using one-sided limits, why $\lim_{x\to 0}\frac{1}{x^2}$ exists as an infinite limit (both one-sided limits agree), while $\lim_{x\to 0}\frac{1}{x}$ does not exist even as an infinite limit.

For $\frac{1}{x^2}$: as $x\to 0^+$ or $x\to 0^-$, $x^2>0$ always, so both one-sided limits are $+\infty$ — they agree, so we say $\lim_{x\to 0}\frac{1}{x^2}=+\infty$. For $\frac{1}{x}$: as $x\to 0^+$, $\frac{1}{x}\to+\infty$, but as $x\to 0^-$, $\frac{1}{x}\to-\infty$ — the one-sided limits disagree (not even both infinite in the same direction), so the limit does not exist in any sense.

13 **[Intermediate]** • **Target: Visualizing a limit graphically**

The graph of $f(x)=\frac{x^2-4}{x-2}$ (which simplifies to $x+2$ with a hole at $x=2$) is shown. Use the graph to estimate $\lim_{x\to 2} f(x)$, and explain why the hole doesn't affect the limit's value.

The graph approaches the height 4 from both sides as $x\to 2$, even though the function is undefined exactly at $x=2$ (the hole). A limit only depends on the function's behavior NEAR the point, not its value (or lack of one) AT the point, so $\lim_{x\to 2} f(x)=4$ regardless of the hole.

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14 [Intermediate] • Target: The formal idea of a limit (intuitive)

Explain, in your own words, what it means for $\lim_{x \to a} f(x) = L$, without using formal epsilon-delta language.

It means that as x gets arbitrarily close to a (from either side, without necessarily equaling a), the function's output values $f(x)$ get arbitrarily close to the single number L.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Limit laws • Cross-Lesson: 4.01 Circles and Parabolas

Given $\lim_{x \to 3} f(x) = 5$ and $\lim_{x \to 3} g(x) = -2$, use the limit laws to evaluate:

a $\lim_{x \to 3} [f(x) + g(x)]$
$= 5 + (-2) = 3$

b $\lim_{x \to 3} [f(x) \cdot g(x)]$
$= 5(-2) = -10$

c $\lim_{x \to 3} \frac{f(x)}{g(x)} = \frac{5}{-2} = -\frac{5}{2}$

(a) $= 5 + (-2) = 3$; (b) $= 5(-2) = -10$; (c) $= \frac{5}{-2} = -\frac{5}{2}$

16 [Challenging] • Target: Limits involving piecewise functions • Cross-Lesson: 4.02 Ellipses

A function is defined by $f(x) = x^2$ for $x < 1$ and $f(x) = 3x - 1$ for $x \geq 1$. Find $\lim_{x \to 1^-} f(x)$, $\lim_{x \to 1^+} f(x)$, and state whether $\lim_{x \to 1} f(x)$ exists.

$\lim_{x \to 1^-} f(x) = 1^2 = 1$. $\lim_{x \to 1^+} f(x) = 3(1) - 1 = 2$. Since the one-sided limits disagree ($1 \neq 2$), $\lim_{x \to 1} f(x)$ does NOT exist.

17 [Challenging] • Target: Limit laws • Cross-Lesson: 4.03 Hyperbolas

On the EST II Math exam, Advanced Algebra and Functions together account for 40% of the questions. In a 50-question test, if a student targets an accuracy rate of 90% in these two high-yield domains, how many questions from these domains can the student afford to miss at most?

Step 1: Find total questions from Advanced Algebra and Functions: $50 \times 0.40 = 20$ questions. Step 2: Calculate 90% accuracy target: $20 \times 0.90 = 18$ questions correct. Step 3: Find maximum allowed missed questions: $20 - 18 = 2$ questions.

18 [Challenging] • Target: Limits involving piecewise functions • Cross-Lesson: 5.01 Vectors: Components and Operations

A function is defined piecewise: $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$. What is the value of $f(-3) + f(2)$?

Evaluate each term according to its interval condition: For $x = -3 < 0$: $f(-3) = (-3)^2 = 9$. For $x = 2 \geq 0$: $f(2) = 2(2) + 1 = 5$. Sum: $f(-3) + f(2) = 9 + 5 = \mathbf{14}$.

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19 [Challenging] • Target: Evaluating limits • Cross-Lesson: 5.02 The Dot Product

If $f(x) = x^2 - 1$ and $g(x) = \sqrt{x+2}$, find (a) $f(g(3))$, (b) the domain of $g \circ f$.

(a) $g(3) = 5$; $f(5) = 5 - 1 = 4$. (b) $g(f(x)) = \sqrt{x^2 - 1 + 2} = \sqrt{x^2 + 1}$; always real, domain: all reals.

20 [Challenging] • Target: One-sided limits and limits at infinity • Cross-Lesson: 5.03 Parametric Equations

If $\sin\theta = -\frac{4}{5}$ and θ lies in Quadrant III, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

Using the Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$: $\left(-\frac{4}{5}\right)^2 + \cos^2\theta = 1 \Rightarrow \frac{16}{25} + \cos^2\theta = 1 \Rightarrow \cos^2\theta = \frac{9}{25}$. Since θ is in Quadrant III, cosine is negative: $\cos\theta = -\frac{3}{5}$. Tangent is positive in Quadrant III: $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{-4/5}{-3/5} = \frac{4}{3}$. Double-angle formula for sine.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating limits

A function is one-to-one if:

a. Every output repeats

b. Passes the vertical line test only

c. **Passes the horizontal line test** ✓

d. Is a polynomial

Horizontal line test confirms 1-1.

22 [Foundational] • Target: One-sided limits and limits at infinity

[Transformation Trial 1] Identify the matrix representing a counterclockwise rotation by 360° in $\mathbb{R}^2$.

a. $(-1, 0; 0, 1)$

b. $(0, 1; 1, 0)$

c. $(1, 0; 0, 1)$ ✓

d. $(1, -1; 1, 1)$

The 2D rotation matrix is $(\cos\theta, -\sin\theta; \sin\theta, \cos\theta)$. For $\theta = 360^\circ$, this evaluates to $(1, 0; 0, 1)$.

23 [Foundational] • Target: More limit computations

Eliminate the parameter to find y as a function of x: $x = 1t - 2$, $y = -2t^2$.

a. $y = -1(x+2)^2$

b. $y = -2(x+2)^2$ ✓

c. $y = 2(x+2)^2$

d. $y = -2(x-2)^2$

From $x = 1t - 2$, $t = \frac{x+2}{1}$. Substitute into $y = -2t^2$: $y = -2\left(\frac{x+2}{1}\right)^2 = -2(x+2)^2$.

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24 [Foundational] • Target: Continuity and limits together

Find the component form of vector $\mathbf{u}$ directed from $(9, 12)$ to $(20, -2)$.

a. $\langle 12, -14 \rangle$

b. $\langle -11, 14 \rangle$

c. $\langle 11, -14 \rangle$ ✓

d. $\langle 11, 14 \rangle$

$\mathbf{u} = \langle (20) - (9), (-2) - (12) \rangle = \langle 11, -14 \rangle$.

25 [Foundational] • Target: Limits from graphs and tables

The function $f(x) = \frac{1}{x-4}$ is undefined at one value of x. Find this value.

a. x=0

b. x=-4

c. x=5

d. **x=4** ✓

The denominator $x - 4$ is zero when $x = 4$, so f is undefined there.

26 [Foundational] • Target: Visualizing a limit graphically

A piecewise function is defined by $f(x) = 4x + 2$ for $x < -3$, and $f(x) = 3x - 4$ for $x \geq -3$. Find $f(-4)$.

a. -15

b. -17

c. **-14** ✓

d. -13

Since $-4 < -3$, use the first piece: $f(-4) = 4(-4) + 2 = -14$.

27 [Foundational] • Target: The formal idea of a limit (intuitive)

Let $f(x) = 3x + 2$ and $g(x) = 1/3x - 2/3$. Are f and g inverse functions?

a. No, f and g are not inverses

b. **Yes, f and g are inverses** ✓

c. Cannot be determined

d. Only if $x = 0$

Check $f(g(x))$: this simplifies to x, so they are inverses.

28 [Foundational] • Target: Limit laws

The function $f(x) = x - e^x$ has exactly one critical point. Find it.

a. $x = e$

b. $x = 1$

c. $x = -1$

d. $x = 0$ ✓

$f'(x) = 1 - e^x = 0 \Rightarrow e^x = 1 \Rightarrow x = 0$. Maximum.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Limits involving piecewise functions**

What does multiplicity 3 mean for a root graphically?

a. The graph touches but doesn't cross

b. The root appears 3 times in distinct places

c. The graph has a cusp

d. **The graph crosses with an inflection-like shape at the root ✓**

Odd multiplicity ≥3: crosses with a flattened 'S' curve.

30 **[Intermediate]** • **Target: Evaluating limits**

Prove: a polynomial of odd degree with real coefficients has at least one real root.

a. **By IVT: $P(x) \to +\infty$ and $P(x) \to -\infty$ (or vice versa) as $x \to \pm\infty$, so it must cross zero ✓**

b. By Vieta's formulas

c. By the Rational Root Theorem

d. Not always true

Sign change at infinity → IVT guarantees crossing zero.

31 **[Intermediate]** • **Target: One-sided limits and limits at infinity**

End behaviour of $f(x) = -2x^5$ as $x \to +\infty$:

a. $+\infty$

b. $-\infty$ ✓

c. 0

d. 2

Odd degree, negative leading coefficient → $-\infty$.

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32 [Intermediate] • Target: More limit computations

Find the determinant of the 2×2 coefficient matrix $A = (4, 2; 3, 4)$.

a. 22

b. **10** ✓

c. 13

d. 6

$\det(A) = ad - bc = (4)(4) - (2)(3) = 16 - 6 = 10$.

33 [Intermediate] • Target: Continuity and limits together

Which transformation shifts $f(x)$ right by 3?

a. $f(x + 3)$

b. $f(x - 3)$ ✓

c. $f(x) + 3$

d. $f(x) - 3$

Horizontal shift right: replace x with $x - 3$.

34 [Intermediate] • Target: Limits from graphs and tables

If $f : \mathbb{R} \to \mathbb{R}$ is both even and odd, then:

a. $f(x) = 0$ **for all** x ✓

b. $f(x) = x^2$

c. $f(x) = x$

d. $f(x) = 1$

Even+odd: $f(x) = -f(x) \Rightarrow f(x) = 0$.

35 [Intermediate] • Target: Visualizing a limit graphically

Determine if $f(x) = \frac{x^3}{x^2 + 1}$ is even, odd, or neither.

a. Neither

b. Constant

c. **Odd** ✓

d. Even

$f(-x) = \frac{-x^3}{x^2 + 1} = -f(x)$: odd.

36 [Intermediate] • Target: The formal idea of a limit (intuitive)

Find $f^{100}(x)$ (100th iterate) of $f(x) = x + 1$.

a. $x + 100$ ✓

b. $x + 2^{100}$

c. $100x$

d. x^{100}

$f^n(x) = x + n$. So $f^{100}(x) = x + 100$.

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37 [Intermediate] • Target: Limit laws

$\tan\theta = \frac{\text{opp}}{\text{adj}}$. In 3-4-5 triangle with opp=3,adj=4:

a. $\frac{3}{4}$ ✓

b. $\frac{4}{3}$

c. $\frac{3}{5}$

d. $\frac{5}{3}$

Applying Unit 3 · Trigonometric and Polar Functions principles, $\tan = \frac{3}{4}$ Therefore, the verified correct option is $\frac{3}{4}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Limits involving piecewise functions • Cross-Lesson: 5.01 Vectors: Components and Operations

A committee of 4 people is to be selected at random from a group of 10 candidates consisting of 4 mathematicians, 3 physicists, and 3 engineers. Given that the selected committee contains at least one person from each of the three professions, what is the probability that exactly 2 mathematicians are on the committee?

a. $\frac{1}{3}$

b. $\frac{9}{35}$

c. $\frac{3}{7}$ ✓

d. $\frac{3}{5}$

To find the conditional probability, we determine the number of committee selections that satisfy the given condition (the reduced sample space) and the number of those selections that contain exactly 2 mathematicians. 1. **Analyze the condition:** The committee must have 4 members chosen from 3 professions with at least one member from each profession. By the Pigeonhole Principle, the members must be partitioned as $2 + 1 + 1$. Thus, exactly one profession will have 2 representatives, and the remaining two profess.

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39 [Challenging] • Target: Evaluating limits • Cross-Lesson: 5.02 The Dot Product

The function $f(x) = a\cos(bx) + d$, where $a, b,$ and d are positive constants, has a period of π, a maximum value of 7, and satisfies $f(\frac{\pi}{6}) = 4$. What is the value of $f(\frac{\pi}{3})$?

a. -2 ✓

b. 1

c. 4

d. -5

We determine the unknown parameters $a, b,$ and d step by step: 1. **Find b using the period:** The period of $f(x) = a\cos(bx) + d$ is given by $\frac{2\pi}{b}$. Setting this equal to π: $\frac{2\pi}{b} = \pi \Rightarrow b = 2$ 2. **Use the maximum value to relate a and d:** Since $a > 0$, the maximum value occurs when $\cos(bx) = 1$: $a(1) + d = 7 \Rightarrow a + d = 7$ 3. **Use the given point $f(\frac{\pi}{6}) = 4$:** Substitute $b = 2$ and $x = \frac{\pi}{6}$.

40 [Challenging] • Target: One-sided limits and limits at infinity • Cross-Lesson: 5.03 Parametric Equations

The polynomial $P(x) = 2x^3 + ax^2 + bx - 6$ has a factor of $(x - 2)$. When $P(x)$ is divided by $(x + 1)$, the remainder is -12. What is the remainder when $P(x)$ is divided by $(x - 3)$?

a. 18

b. **24** ✓

c. 30

d. 48

We use the Factor Theorem and the Remainder Theorem: 1. Since $(x - 2)$ is a factor of $P(x)$, we have $P(2) = 0$: $P(2) = 2(2)^3 + a(2)^2 + b(2) - 6 = 16 + 4a + 2b - 6 = 4a + 2b + 10 = 0$ $2a + b = -5 \quad - - -$ (Equation 1) 2. Since the remainder when divided by $(x + 1)$ is -12, we have $P(-1) = -12$: $P(-1) = 2(-1)^3 + a(-1)^2 + b(-1) - 6 = -2 + a - b - 6 = a - b - 8 = -12$ $a - b = -4 \quad - - -$ (Equation 2) 3. Add Equation 1 and Equation 2 to eliminate b.

41 [Challenging] • Target: More limit computations • Cross-Lesson: 6.01 Polar Coordinates and Conversions

The polynomial $P(x) = x^4 + ax^3 - 7x^2 + bx - 12$ has zeroes at $x = -3$ and $x = 2$, where a and b are real constants. What is the value of $a + b$?

a. -18

b. -2

c. 18 ✓

d. 22

By the Factor Theorem, if $x = 2$ and $x = -3$ are zeroes of $P(x)$, then $P(2) = 0$ and $P(-3) = 0$. 1. Substitute $x = 2$: $P(2) = (2)^4 + a(2)^3 - 7(2)^2 + b(2) - 12 = 0$ $16 + 8a - 28 + 2b - 12 = 0 \Rightarrow 8a + 2b - 24 = 0 \Rightarrow 4a + b = 12$ (Equation 1) 2. Substitute $x = -3$: $P(-3) = (-3)^4 + a(-3)^3 - 7(-3)^2 + b(-3) - 12 = 0$ $81 - 27a - 63 - 3b - 12 = 0 \Rightarrow -27a - 3b + 6 = 0 \Rightarrow 9a + b = 2$ (Equation 2) 3. Solve the system of linear equations: Subt.

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42 [Challenging] • Target: Continuity and limits together • Cross-Lesson: 6.02 Polar Equations and Graphs

Consider the sequence a_n defined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 + 4}{2a_n}$ for $n \geq 1$. If the sequence converges to a limit L, find the value of $L^2 - 2L$.

a. **1** ✓

b. 2

c. 3

d. 4

The sequence is defined by the recurrence relation $a_{n+1} = \frac{a_n^2 + 4}{2a_n}$. If the sequence converges to a limit L, then as $n \to \infty$, we have $a_{n+1} \to L$ and $a_n \to L$. Substituting L into the recurrence relation, we get: $L = \frac{L^2 + 4}{2L}$ Multiplying both sides by $2L$ (assuming $Leq0$), we have: $2L^2 = L^2 + 4$ $L^2 = 4$ This gives $L = 2$ or $L = -2$. Let's check the first few terms of the sequence to determine the sign of the limit. $a_1 = 3$..

43 [Challenging] • Target: Limits from graphs and tables • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

The function f is defined for $x \in \mathbb{R}, xeq3$ by $f(x) = \frac{2x+1}{x-3}$. Which of the following is an expression for the inverse function $f^{-1}(x)$, where $xeq2$?

a. $\frac{3x+1}{x-2}$ ✓

b. $\frac{x-3}{2x+1}$

c. $\frac{3x-1}{x+2}$

d. $\frac{x+1}{2x-3}$

To find the inverse function $f^{-1}(x)$, set $y = f(x)$ and rearrange to express x in terms of y: $y = \frac{2x+1}{x-3}$ Multiply both sides by $x - 3$: $y(x-3) = 2x + 1$ $yx - 3y = 2x + 1$ Collect all terms containing x on one side and constant/independent terms on the other: $yx - 2x = 3y + 1$ Factor out x: $x(y-2) = 3y + 1$ Divide by $y - 2$: $x = \frac{3y+1}{y-2}$ Interchanging the variables gives: $f^{-1}(x) = \frac{3x+1}{x-2}$ Common Misconceptions: - Choice.

44 [Challenging] • Target: Visualizing a limit graphically • Cross-Lesson: 7.02 Logarithms and Their Properties

The bar graph below shows the number of students enrolled in each of four elective courses at a high school. If one student is selected at random from these four courses, what is the probability that the selected student is enrolled in Music?

a. $\frac{7}{50}$

b. $\frac{7}{20}$ ✓

c. $\frac{7}{13}$

d. $\frac{13}{20}$

To find the probability that a randomly chosen student is enrolled in Music, divide the number of students enrolled in Music by the total number of students enrolled across all four courses. 1. Find the total number of students: $\text{Total} = 8 + 14 + 6 + 12 = 40$ 2. Find the number of students in Music: $n(\text{Music}) = 14$ 3. Calculate the probability: $P(\text{Music}) = \frac{14}{40} = \frac{7}{20}$ Common misconceptions: - $\frac{7}{13}$ computes the odds in favor (ratio of students in Music to studen.

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45 [Challenging] • Target: The formal idea of a limit (intuitive) • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

The first three terms of a geometric progression are 32, 24, and 18. Find the sum to infinity, S_∞, of the progression.

a. 128 ✓

b. 96

c. 74

d. 64

The first term of the geometric progression is $a = 32$. The common ratio r is found by dividing any term by the preceding term: $r = \frac{24}{32} = \frac{3}{4}$ Since $|r| = \left|\frac{3}{4}\right| < 1$, the geometric progression converges and the sum to infinity exists. The sum to infinity is given by: $S_\infty = \frac{a}{1-r}$ Substituting $a = 32$ and $r = \frac{3}{4}$: $S_\infty = \frac{32}{1-\frac{3}{4}} = \frac{32}{\frac{1}{4}} = 32 \times 4 = 128$ Common misconceptions: - Choice B (96) arises fro.

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